DEBADRI SANYAL

United States (Open to Relocation) | STEM-OPT Eligible (EAD, Work Authorized Nationwide)
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PROFESSIONAL SUMMARY

Data Scientist with 3+ years designing and deploying end-to-end ML, agentic AI, and LLM systems that drive business solutions for Fortune 500 clients across operations, manufacturing, and healthcare. Hands-on with AWS architecture, NLP frameworks, agentic observability, RAG pipelines, and production-grade data engineering. Proven track record leading cross-functional projects and managing client engagements through full delivery lifecycle.

EDUCATION

Purdue University, Daniels School of Business — West Lafayette, IN
MS Business Analytics & Information Management, Data Science Specialization (STEM), GPA: 3.8/4.0 — Aug 2025 – Dec 2026
Relevant Coursework: Machine Learning, Deep Learning, Applied Statistics, Python for Analytics, Data Mining, Experimentation & Causal Inference, GenAI & LLMs, Data Visualization, Data Storytelling

Vellore Institute of Technology — Tamil Nadu, India
B.Tech, Electronics & Instrumentation Engineering — Jul 2018 – Jun 2022

PROFESSIONAL EXPERIENCE

Cummins — West Lafayette, IN
AI & Data Science Lead, The Data Mine, Purdue — *Sep 2026 – Present*
- Leading 4 AI project divisions for Cummins' enterprise Analytics and Artificial Intelligence group, overseeing agentic AI observability, enterprise CI/CD with persistent-memory agents, asset-based diagnostics optimization, and AI-enabled compliance automation across 150+ students and 20+ corporate stakeholders on Databricks and Azure.
- Architecting an agentic observability cockpit to track AI agent token usage, accuracy, and performance across Cummins' digital core (Databricks, Snowflake, Palantir, Neo4j), and building persistent-memory agents using LangGraph, RAG (Mosaic AI Vector Search), and MLflow evaluation to enable multi-turn context retention, reducing agent development time by 30-40%.

Amazon — Bellevue, WA
Data Scientist Intern — *Jun 2026 – Aug 2026*
- Engineered a capacity recommendation engine for planners who manually processed 150,000+ station-weeks across 20+ disconnected systems with inconsistent and unauditable decisions, achieving 90.5% agreement across 10,000+ backtested weeks on AWS (Lambda, S3, DuckDB, DynamoDB) scaled to US, Canada, and Europe.
- Codified planner mental models into a decision tree (20+ nodes, 25+ scenarios) and built an XGBoost imitation model replicating planner behavior (AUC 0.92). Introduced a Decision Quality Score to benchmark decisions, surfaced divergence points, and optimized thresholds to maximize decision quality.

Google — United States (Remote)
Data Science Consultant, Industry Practicum, Purdue — *Jan 2026 – May 2026*
- Built a creator-side content recommender for YouTube creators whose analytics showed raw view counts with survivorship bias hiding breakout opportunities for smaller channels, designing a custom Outlier Multiplier for size-agnostic content detection with AI-powered strategy briefs.
- Engineered BERTopic semantic clustering on YouTube Data API v3 to surface hidden content trends, integrated OpenAI and Gemini APIs for strategy synthesis, and deployed via Streamlit with Docker and GitHub Actions CI/CD pipeline.

Deloitte — Hyderabad, India
Senior Business Analyst — *Sep 2022 – Aug 2025*
- Delivered executive dashboards for Fortune 500 clients in financial services, insurance, and healthcare who lacked unified reporting across fragmented systems with 10M+ daily records, reducing reporting latency by 35% on $4M+ annualized engagements and earning the Deloitte Rising Star Award.
- Developed predictive fraud detection models for insurance clients using gradient-boosted classifiers on claims data (91% precision), built Tableau/Power BI dashboards on Snowflake with advanced SQL (CTEs, window functions), and translated data patterns into actionable recommendations for VP/Director leadership.

Gyanscore Pvt. Ltd. — Delhi, India
Business Analyst Intern — *May 2021 – Aug 2021*
- Built a predictive analytics pipeline for engineering teams who investigated sensor failures reactively with no forecasting capability, developing a regression model to predict upcoming fault sensors from current performance metrics across 100K+ IoT records, replacing manual investigation with automated early-warning workflows.

PROJECTS

Financial Fraud Detection at Scale
- Replaced legacy rule-based fraud systems that missed evolving transaction patterns with classification ML models, achieving AUC-ROC 0.9460 on 590K+ transactions using XGBoost + LightGBM with 200+ behavioral features and SMOTE resampling.

Corporate Bankruptcy Prediction
- Predicted corporate bankruptcy for lenders lacking early default signals, building gradient-boosted classifiers with financial ratio engineering across 28 iterations to flag at-risk companies.

AI Factuality Detection | 1st Place Midwest, DataCamp Data4Good
- Detected hallucinated claims in LLM-generated content to combat AI-driven misinformation, achieving 93.54% Macro-AUC using ensemble NLP classifiers with transformer-based feature extraction.

TECHNICAL SKILLS

- **Languages & ML Frameworks:** Python (Pandas, NumPy, Scikit-Learn, PyTorch, TensorFlow), Advanced SQL, R, XGBoost, LightGBM, Hugging Face Transformers, Statsmodels, Git, Linux (Shell).
- **ML, AI & Experimentation:** Deep Learning, NLP, Agentic AI, LLMs (GPT-4, BERTopic, RAG), LangGraph, GenAI, Causal Inference, A/B Testing, Bayesian Optimization, MLflow, SageMaker, Experiment Tracking.
- **Cloud, Data & BI:** AWS (S3, Glue, Lambda, SageMaker), Databricks, Snowflake, Airflow, Docker, Tableau, Power BI, Data Storytelling, Executive Readouts, Cross-Functional Collaboration, Agile & Scrum.
- **Awards & Certifications:** 1st Place Future Edelman Award (Purdue), Deloitte Rising Star, Prudential SPOT Award, DataCamp DS Associate, Azure AI-900, Databricks Fundamentals.